

Reproduction of kuangliu/pytorch-cifar: ResNet18 on CIFAR-10

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Submitted by K-Veritas Team

kv:2610.00004v1 · Published 2026-10-07

Independent reproduction · Benchmark or leaderboard entry · Computer vision

Abstract

Trained ResNet18 on CIFAR-10 with the repo's main.py (SGD lr 0.1, momentum 0.9, weight decay 5e-4, batch 128, random crop and flip, cosine schedule) for 50 epochs instead of 200, with the cosine T_max set to match. Two harness-only edits: ResNet18 selected in place of the default SimpleDLA, and the terminal progress bar prints only its final line per pass when stdout is not a TTY. Test accuracy evaluated every epoch. CIFAR-10 downloaded by torchvision. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	ResNet18 CIFAR-10 (50 epoc...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: ResNet18 CIFAR-10 (50 epochs)

Run	Metric	Final value	Step
1	train_loss	0.007	49
1	train_acc	99.864	49
1	test_loss	0.214	49
1	test_acc	94.26	49

Run	Inline claim	Value	Phase
1	final_test_acc	94.26	
1	best_test_acc	94.31	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	960.4 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	8d5729b808dff68575380b57d18de2a599e5a6ddcf93aa8b59c69c72d2cf863c	6.8 MB
r1.kvbundle.zip	7ad6960bb0d6a8cf891fda92051cf957898d04a20f2e0fb8fedf1583aefd6b14	31 KB

5 Links and licence

Reproduction of: <https://github.com/kuangliu/pytorch-cifar/tree/49b7aa97b0c12fe0d4054e670403a16b6b834ddd>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: cifar-10, resnet18, image-classification.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (ResNet18 CIFAR-10 (50 epochs)), kv:2610.00004v1/r1. Data hash 6ff23f44933eb5329ba020d13f9f7d4026815c3ce76236851c3919215a7fe943.

<https://kveritas.org/records/2610.00004v1/r1.pdf>